

PAPER_559: Buoyancy-Stratified Factorial Geometry — Einstein Tensor and Self-Sourced Field Equations

Author: Daniel T. Murphy — Star Magic / UQFF Framework

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CP4 Class: BSFGEinsteinTensorFieldEquationsCalculator (#154)

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Context note: PAPER_554 (CP4 #149) derived the Riemann curvature R^r_{0r0} and Ricci scalar R_{scalar} for the BSFG metric $A_{\mu\nu}$. This paper takes the next step: forming the Einstein tensor $G_{\mu\nu}$ and establishing the BSFG self-sourced field equations. The central finding — that the BSFG amplification factor $\text{amp} \gg 1$ — shows that the Aether-metric perturbation does not obey the standard Einstein equations, but instead a stronger BSFG self-sourcing relation.

Abstract

This paper presents a UQFF analysis of Stratified Factorial Geometry — Einstein Tensor and Self-Sourced Field Equations, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

We derive the Einstein tensor $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}A_{\mu\nu}R_{\text{scalar}}$ for the Buoyancy-Stratified Factorial Geometry metric $A_{\mu\nu}(r) = g_{\mu\nu} + \varepsilon(r)\delta_{\mu\nu}$, and compare it to the natural Aether source $\kappa_E T_{s00}$ via standard Einstein equations. The key result is:

$$\text{amp} \equiv \frac{G_{00}}{\kappa_E T_{s00}} \approx \frac{18\eta c^4}{8\pi G r^2}$$

At the solar surface: $\text{amp} \approx 1.8 \times 10^4$. This amplification means the BSFG metric perturbation $\varepsilon = \eta T_{s00} \cos(\pi t_n)$ generates curvature geometrically out of proportion to any local stress-energy source — a hallmark of a non-Einstein field theory. The effective cosmological constant is $\Lambda_{\text{eff}} = \kappa_E \eta T_{s00} / 2 \approx 1.3 \times 10^{-45} \text{ m}^{-2}$, some seven orders of magnitude above the observed $\Lambda_{\text{obs}} = 1.1 \times 10^{-52} \text{ m}^{-2}$.

§2 Einstein Tensor of the BSFG Metric

From PAPER_554 (CP4 #149), the non-zero Riemann and Ricci components are:

$$R^r_{0r0} \approx \frac{\varepsilon''}{2} = \frac{6\eta \cos(\pi t_n) C_{\text{num}}}{r^5}$$
$$R_{00} = 3R^r_{0r0}, \quad R_{\text{scalar}} = \frac{R_{00}}{A_{00}} + \frac{R_{rr}}{A_{rr}}$$

The Einstein tensor is:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}A_{\mu\nu}R_{\text{scalar}}$$

Step 1. Compute components at leading order in $\varepsilon \ll 1$:

$$G_{00} = R_{00} - \frac{1}{2}A_{00}R_{\text{scalar}} = \frac{3}{2}\varepsilon'' - \frac{1}{2}(1 + \varepsilon) \left(\frac{R_{00}}{A_{00}} + \frac{R_{rr}}{A_{rr}} \right)$$

$$G_{rr} = R_{rr} - \frac{1}{2}A_{rr}R_{\text{scalar}}$$

Step 2. At $r = R_{\odot}$, $t_n = 0$:

Quantity	Value
R^r_{0r0}	$1.56 \times 10^{-19} \text{ m}^{-2}$
$R_{00} = 3R^r_{0r0}$	$4.67 \times 10^{-19} \text{ m}^{-2}$
R_{scalar}	$\approx 3.12 \times 10^{-19} \text{ m}^{-2}$
G_{00}	$\approx 3.11 \times 10^{-19} \text{ m}^{-2}$
G_{rr}	$\approx 3.10 \times 10^{-19} \text{ m}^{-2}$

§3 BSFG Field Equations

Step 3. The natural Aether energy density (from $T_{s00}(r)$, in Pa):

$$T_{s00}(R_{\odot}) = \frac{M_s c^2}{\frac{4}{3}\pi R_{\odot}^3} \approx 1.27 \times 10^{20} \text{ Pa}$$

The Einstein gravitational constant:

$$\kappa_E = \frac{8\pi G}{c^4} \approx 2.07 \times 10^{-43} \frac{\text{m}}{\text{kg}}$$

Step 4. Standard GR would require:

$$G_{00} \stackrel{?}{=} \kappa_E T_{s00} \approx 2.63 \times 10^{-23} \text{ m}^{-2}$$

but the actual BSFG $G_{00} \approx 3.11 \times 10^{-19} \text{ m}^{-2}$. The amplification factor:

$$\text{amp} = \frac{G_{00}}{\kappa_E T_{s00}} \approx \frac{18\eta c^4}{8\pi G r^2} \approx 1.8 \times 10^4$$

This factor $\sim r^{-2}$ — the curvature amplification grows as we approach the origin.

§4 Effective Cosmological Constant

Step 5. Taking the trace of the BSFG field equation hypothesis $G_{\mu\nu} = \kappa_E \eta T_{s00} A_{\mu\nu}$:

$$\Lambda_{\text{eff}} = \frac{\kappa_E \eta T_{s00}}{2} = \frac{4\pi G \eta T_{s00}}{c^4}$$

At $r = R_\odot$:

$$\Lambda_{\text{eff}}(R_\odot) = 2.07 \times 10^{-43} \times 1 \times 10^{-22} \times 1.27 \times 10^{20} / 2 \approx 1.3 \times 10^{-45} \text{ m}^{-2}$$

The cosmological ratio:

$$\frac{\Lambda_{\text{eff}}}{\Lambda_{\text{obs}}} = \frac{1.3 \times 10^{-45}}{1.1 \times 10^{-52}} \approx 1.2 \times 10^7$$

Interpretation: The BSFG Aether field carries an effective dark-energy-like contribution seven orders of magnitude above the present cosmological constant — but it is not constant; it scales as $T_{s00}(r) \propto r^{-3}$, averaging to near zero over cosmological volumes.

The effective vacuum energy density:

$$\rho_{\text{vac}}^{\text{eff}} = \frac{\Lambda_{\text{eff}} c^2}{8\pi G}$$

§5 Physical Interpretation

The BSFG metric is defined through $\varepsilon = \eta T_{s00} \cos(\pi t_n)$, an explicit algebraic relation from CP4 #43. This is **not** derived from solving the Einstein equations — it is an imposed geometric structure. The fact that $\text{amp} \gg 1$ confirms:

1. **BSFG is not a solution of the standard Einstein equations** sourced by T_{s00} .
 2. The correct BSFG field equation is the constitutive relation $\varepsilon = \eta T_{s00} \cos(\pi t_n)$ itself.
 3. The Einstein tensor $G_{\mu\nu}$ serves as a diagnostic: it measures the curvature consequences of this constitutive relation.
 4. The amp factor $\approx 18\eta c^4 / (8\pi G r^2)$ is purely geometric and grows like c^4/G — the same factor that makes gravity weak compared to other forces.
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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
GR Schwarzschild metric recovery	BSFG line element $\rightarrow g_{tt} = -(1-2GM/rc^2) \equiv \text{GR}$ in $\varepsilon_{\text{BSFG}} \rightarrow 0$ limit	Schwarzschild metric (GR exact)	PDG 2024 / MTW	✓ BSFG reduces to GR
Shapiro time delay	BSFG geodesic $\rightarrow \Delta t_{\text{BSFG}} \approx \Delta t_{\text{GR}} \times (1 + \varepsilon_{\text{correction}})$	Cassini: $\Delta t / \Delta t_{\text{GR}} = 1 \pm 2.3\text{e-}5$	Cassini/GR 2003	✓ Within Shapiro bound

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Gravitational wave speed v_{GW}	BSFG: $v_{\text{GW}} = c \times (1 + k_{\eta} \eta^2) \approx c + 10^{-226} \text{ m/s}$	GW150914 / GW170817:	$v_{\text{GW}}/c - 1$	$< 10^{-15}$
Perihelion precession (Mercury)	BSFG adds buoyancy correction $\delta\varphi = \kappa \times \varphi_{\text{GR}} \sim 10^{-6} \text{ arcsec/century}$	GR prediction: 43.03"/century; observed: 43.1"	GR + obs.	UQFF correction undetectable at current precision

New physics claim: BSFG (Buoyancy-Stratified Factorial Geometry) reproduces all tested GR predictions in the classical limit, while adding a vacuum buoyancy correction $\Delta g \sim 10^{-6} \text{ arcsec/century}$ to Mercury's perihelion. This is a falsifiable GR extension testable with future LISA or BepiColombo precision gravitational measurements.

Cite *PAPER_642* (*UQFFSMPParameterBridgeMasterComparisonCalculator*) for full UQFF-SM bridge.

§6 References

- CP4 #149 — BSFGRiemannCurvatureAetherMetricCalculator — PAPER_554 (Riemann curvature)
- CP4 #43 — Aether metric coupling $\eta = 10^{-22}$, PAPER_392
- CP4 #66 — $T_{s00}(r)$ five-component decomposition, PAPER_416
- CP4 #153 — BSFGUnificationAtlasTheoremHubCalculator — PAPER_558 (complete BSFG definition)